Dear Quillstone AI Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. As a Data Scientist moving into Machine Learning Engineering, I have built and operated scheduled forecasting and batch-scoring workflows in Python and SQL, and I am keen to bring that experience to machine-learning systems that help merchants get support quickly and without friction.

At Thistlebox, I developed a six-week-ahead recipe-demand forecasting workflow using LightGBM, orchestrated as scheduled notebook jobs in Vertex AI Workbench, with outputs written to BigQuery for operational reporting. The workflow improved WAPE from 23% to 14%, with an estimated 9% reduction in food waste. I also built a weekly XGBoost churn-scoring workflow in BigQuery and validated retention targeting with a holdout, where churn decreased by 2.1 percentage points in the treated group. These projects have given me practical experience in Python and SQL, feature engineering, model evaluation, batch scoring, and orchestration through scheduled ML workflows on GCP.

I have also added pytest unit tests for forecasting feature code, reaching approximately 60% coverage, and built and containerised a Python Streamlit application, writing its Dockerfile; the data platform team handled its deployment to Cloud Run. My MSc in Statistics, including a dissertation on Bayesian hierarchical models, supports my modelling and experimental-design work. I am now looking to develop further as an ML engineer, contributing careful testing, ownership and practical problem-solving as I deepen my production engineering experience.

Quillstone's focus on support automation and its emphasis on team ownership, open sharing and learning from feedback make this role especially appealing to me. I would welcome the opportunity to discuss how my experience with forecasting, churn modelling, feature engineering, batch-scoring workflows and automated testing could contribute to the AI team.

Kind regards,

Tobias Mwangi-Hartley